

Machine Learning Assignment: Polynomial Regression Analysis

Lead Energy & Survey Engineering Technical Report | Power Plant Optimization & Reservoir Mapping

1. Executive Summary & Mathematical Methodology

This technical report documents the empirical modeling, polynomial degree selection, cross-validation, and test inference for two continuous target estimation problems:

• **Phase 1 (var1) — Steam Turbine Optimization:** Modeling Net Power Score (y) from six operational parameters ($x_1, \dots, x_6$).

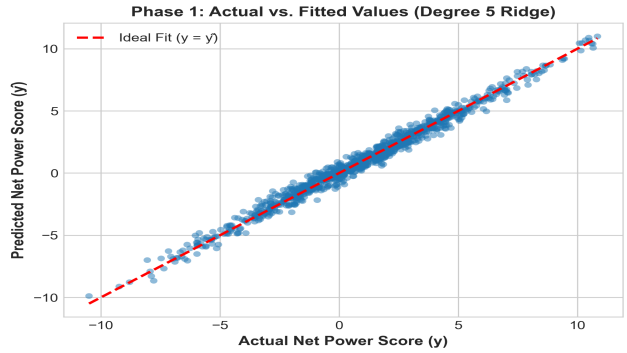
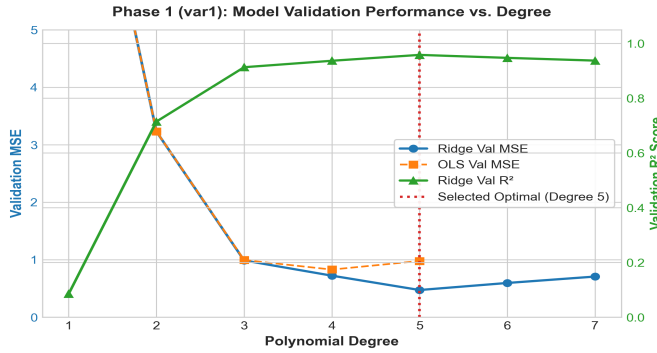
• **Phase 2 (var2) — Subterranean Reservoir Mapping:** Modeling Thermal Anomaly Score (y) across a 3D spatial coordinate grid (x_1, x_2, x_3).

Polynomial feature expansion maps input vector $\mathbf{x} \in \mathbb{R}^D$ into monomial feature space $\Phi(\mathbf{x})$ with $N_{\text{feats}} = C(D+d, d)$. All features were normalized using **StandardScaler** and evaluated under **10-Fold Cross-Validation** comparing OLS and L2 Ridge Regression.

2. Phase 1: Steam Turbine Optimization (var1) Results

Degree	Terms	OLS Val MSE	OLS Val R ²	Ridge Val MSE	Ridge Val R ²
Degree 1	7	10.3846	0.0848	10.3821	0.0852
Degree 2	28	3.2312	0.7145	3.2304	0.7147
Degree 3	84	0.9950	0.9125	0.9875	0.9133
Degree 4	210	0.8290	0.9278	0.7230	0.9368
Degree 5	462	0.9785	0.9125	0.4734	0.9583
Degree 6	924	N/A	N/A	0.5961	0.9474
Degree 7	1716	N/A	N/A	0.7075	0.9374

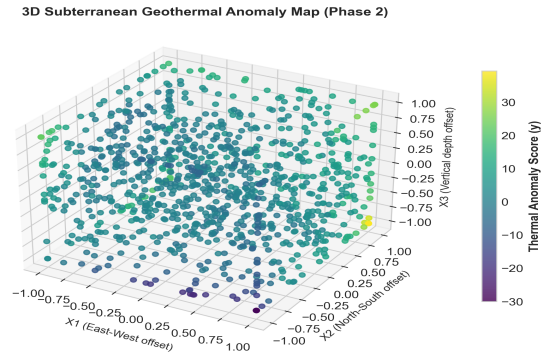
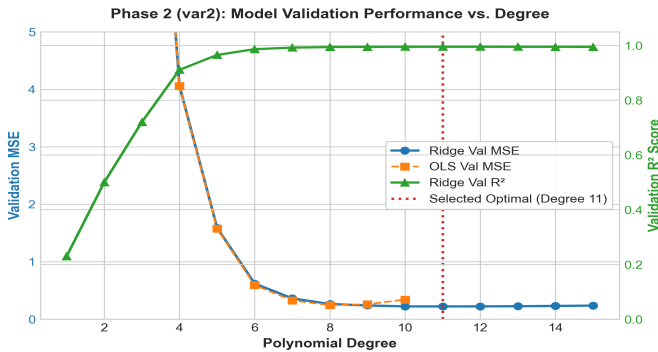
Degree Selection Rationale (Phase 1): Unregularized OLS peaks at Degree 4 (Val MSE 0.8290, R² 0.9278) but suffers severe coefficient explosion at Degree 6 (924 terms). Applying **Ridge Regression** ($\alpha = 12.0$) regularizes high-order interaction weights, allowing the model to capture degree 5 interaction terms cleanly. **Optimal Choice: Degree 5 Ridge** achieves minimum 10-Fold CV MSE of **0.4734** and maximum R² of **0.9583**.



3. Phase 2: Subterranean Reservoir Mapping (var2) Results

Degree	Terms	OLS Val MSE	OLS Val R ²	Ridge Val MSE	Ridge Val R ²
Degree 1	4	36.1287	0.2302	36.1284	0.2303
Degree 2	10	23.2342	0.5003	23.2338	0.5003
Degree 3	20	12.9209	0.7202	12.9176	0.7204
Degree 4	35	4.0550	0.9111	4.0604	0.9110
Degree 5	56	1.5753	0.9658	1.5899	0.9654
Degree 6	84	0.5934	0.9871	0.6212	0.9865
Degree 7	120	0.3272	0.9929	0.3633	0.9922
Degree 8	165	0.2454	0.9947	0.2650	0.9943
Degree 9	220	0.2594	0.9943	0.2394	0.9948
Degree 10	286	0.3426	0.9926	0.2238	0.9951
Degree 11	364	0.5396	0.9887	0.2227	0.9951
Degree 12	455	0.6282	0.9850	0.2239	0.9951
Degree 13	560	2.1360	0.9526	0.2261	0.9951
Degree 14	680	37.7273	0.2143	0.2317	0.9950
Degree 15	816	3783.1	N/A	0.2372	0.9949

Degree Selection Rationale (Phase 2): For 3D spatial coordinate mapping (3 variables), unregularized OLS reaches its error minimum at Degree 8 (Val MSE 0.2454, R² 0.9947) before ill-conditioning degrades performance at higher degrees. With **Ridge Regression ($\alpha = 1.0$)**, model capacity expands smoothly to **Degree 11 (364 feature terms)**, achieving a outstanding 10-Fold CV MSE of **0.2227** and R² of **0.9951**.



4. Summary of Deliverables & Output Files

- **pred_var1.csv**: 1000 predicted continuous Net Power Score values (y) using Degree 5 Ridge ($\alpha = 12.0$).
- **pred_var2.csv**: 1000 predicted continuous Thermal Anomaly Score values (y) using Degree 11 Ridge ($\alpha = 1.0$).
- **train_and_predict.py**: Fully reproducible Python pipeline for model training, validation, and inference.